



GRADUATION PROJECT • 2025 / 2026

QuantWise

An AI-Powered Decision-Support Stock Advisory Platform

Personalised, Risk-Graded Investment Recommendations for Retail Investors

TEAM

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ALIGNED SDGS

9 INDUSTRY, INNOVATION &
INFRASTRUCTURE

10 REDUCED INEQUALITIES

Background

Commission-free apps — Robinhood, eToro, and Egypt's Thndr — have put the stock market in everyone's pocket. The knowledge to use it well did not arrive with them.

~23%

of US equity trading volume is now retail — up from under 15% in 2019

>30%

growth in US retail brokerage accounts between 2019 and 2022

- 01 **The domain.** Decision-support investing — turning prices, forecasts, news sentiment, and risk profiling into a clear "what should I do?" — *not* automated trading.
- 02 **The gap.** That guidance traditionally needs financial expertise or costly advisory services most first-time investors cannot reach.
- 03 **The result.** New investors face information overload and fall back on generic robo-advice or guesswork — leading to avoidable losses.

Sources: FINRA — growth in U.S. retail brokerage participation (2019-2022); Bloomberg Intelligence — retail share of U.S. equity trading volume (2023).

Motivation

"As a Thndr user, I watched friends download the app — then freeze. They had the account, but no idea what to buy or how to start."

That gap is the reason QuantWise exists: to guide everyday retail investors who know nothing about the market toward safe, informed entry.

LEAVING IT IN THE BANK

Returns below inflation

Savings interest often fails to beat the yearly inflation rate — real money quietly loses value over time.

LOW-RISK EQUITY INVESTING

Safer than it looks — higher real returns

Diversified, risk-graded stock exposure has historically outpaced bank savings, with risk kept deliberately low.

Challenges

Four limitations stood out across the research — and became the problems QuantWise set out to tackle.

Quant forecasts alone

Deep-learning and tree models predict price direction well, but carry no sentiment, no risk grading, and no personalization.

→ **Pair forecasts with sentiment + a risk engine**

Sentiment-only models

Improve signal reliability, yet don't produce equity-level, explainable, per-user recommendations.

→ **Fuse sentiment into one graded signal**

Ungrounded LLMs

Generative models can read markets from text, but hallucinate — unsafe for direct financial advice.

→ **Constrain the LLM to pre-validated signals**

Generic robo-advisors

Personalize only at the asset-class level: no per-stock calls, no rationale, static between rebalances.

→ **Per-stock, risk-personalized, explained picks**

The questions we set out to answer

Non-expert retail investors lack guidance that is at once personalised, risk-aware, grounded, and interpretable. We frame that gap as four questions — each with the improvement QuantWise targets.

Q1

How can market-wide ML forecasts become per-user, risk-graded picks a beginner can act on at a glance?

↓ Decision time

Q2

Can a generative LLM personalise advice without inventing prices, tickers, or numbers?

↑ Reliability

Q3

Can quantitative predictions and market sentiment be fused into one trustworthy signal?

↑ Accuracy

Q4

Can personalised AI advice be served instantly and cost-effectively, every trading day?

↓ Latency & cost

OBJECTIVE

What QuantWise sets out to achieve

Make professional-grade, **risk-aware investing** genuinely accessible — so anyone can start with confidence, regardless of their financial background.

◆ Personalised plans

A BUY / SELL / HOLD plan with allocations and plain-language reasons, tailored to each user's risk profile.

◆ Clarity over noise

Turn a daily flood of prices, news, and signals into one confident decision in seconds.

◆ Trustworthy AI

Advice users can rely on — the model only synthesises validated signals; it never forecasts or invents numbers.

◆ Investing for everyone

Help first-time investors beat inflation safely and close the gap with institutional players (SDG 9 & 10).

LSTM for financial market prediction

Fischer & Krauss · European Journal of Operational Research · 2018

MODEL

A Long Short-Term Memory (LSTM) recurrent neural network — a deep sequence model with gated memory cells for time-series.

ARCHITECTURE

Trained on the full S&P 500 constituent history; sequences of past returns feed the LSTM to predict which stocks beat the cross-sectional median next day.

RESULTS

+0.46% / day

Daily returns of ~0.46% before costs (Sharpe $\approx$ 5.8), outperforming random forests, deep nets, and logistic regression.

LIMITATION

Uses price data only — no news sentiment — and the excess returns decay after 2010 as markets grow more efficient.

→ **QuantWise fuses sentiment + risk grading onto the forecast.**

FinBERT: domain-tuned financial sentiment

Araci · [arXiv:1908.10063](#) · 2019

MODEL

BERT — a transformer language model — further pre-trained on a large financial-news corpus, then fine-tuned for sentiment.

ARCHITECTURE

Stacked self-attention encoders produce context-aware embeddings; a classification head labels financial text as positive / negative / neutral.

RESULTS

0.86 accuracy

State of the art on the Financial PhraseBank, beating prior lexicon and ML methods even with little labelled data.

LIMITATION

Classifies sentiment only — it says nothing about price direction or magnitude, so a positive headline can contradict a falling forecast.

→ **QuantWise cross-validates FinBERT against the quant signal.**

LLMs forecasting from financial news

Lopez-Lira & Tang · arXiv:2304.07619 · 2023

MODEL

A large language model (ChatGPT / GPT-class) prompted to read news headlines and rate their sentiment for a stock.

ARCHITECTURE

Zero-shot prompting: each headline becomes an LLM sentiment score, aggregated into a daily signal used to rank stocks long/short.

RESULTS

Significant α

LLM news sentiment showed statistically significant predictive power for next-day returns; the long-short strategy was profitable.

LIMITATION

The model reasons about markets ungrounded — it can hallucinate, carries no user risk profile, and isn't anchored to validated signals.

→ QuantWise constrains the LLM to synthesise pre-graded signals only.

14 US equities · 2010 – 2017

TASK

Regression

OUTPUT

Continuous return → up / down

MODALITY

Numerical time-series

GRANULARITY

Daily OHLCV

FEATURES

One trading day per stock is a record. Raw **Open-High-Low-Close-Volume** prices feed **14 hand-crafted technical indicators** (RSI, MACD, SMA/EMA ratios, momentum, volatility).

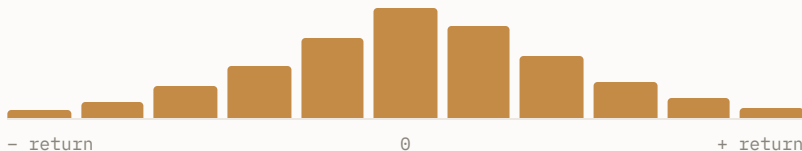
The LSTM reads a **60-day window**; its embedding is stacked with the indicators into a **78-feature** vector for the XGBoost head.

Target — forward return at 30 / 90 / 252 / 365-day horizons (continuous; direction derived as a binary up/down label). Not a classification dataset.

SAMPLE — AAPL, DAILY OHLCV · 2017

DATE	OPEN	HIGH	LOW	CLOSE	VOL
2017-01-03	115.80	116.33	114.76	116.15	28.8M
2017-01-04	115.85	116.51	115.75	116.02	21.1M
2017-01-05	115.92	116.86	115.81	116.61	22.2M
2017-01-06	116.78	118.16	116.47	117.91	31.8M

TARGET DISTRIBUTION — FORWARD RETURN (≈ NORMAL)



~100 US large-caps · 2015 – 2025

TASK	OUTPUT	MODALITY	GRANULARITY
Regression	30-day return → up / down	Numerical time-series	Daily OHLCV

FEATURES & COLLECTION

Manually assembled via the **yfinance** API: daily OHLCV for ~100 diverse US large-caps (2015–2025), deliberately including decliners (BA, INTC, PYPL) to avoid "bull-market survivor" bias.

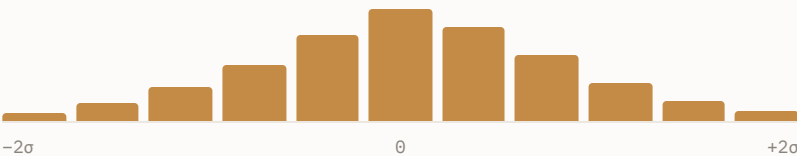
Each ticker is downloaded, features engineered in-code (5 sequential inputs + 14 indicators, 60-day look-back), then split **chronologically 70 / 15 / 15** with train-only target normalisation.

Target — forward 30-day cumulative return, Z-normalised (mean 0.019, std 0.105). Continuous regression; direction derived as up/down.

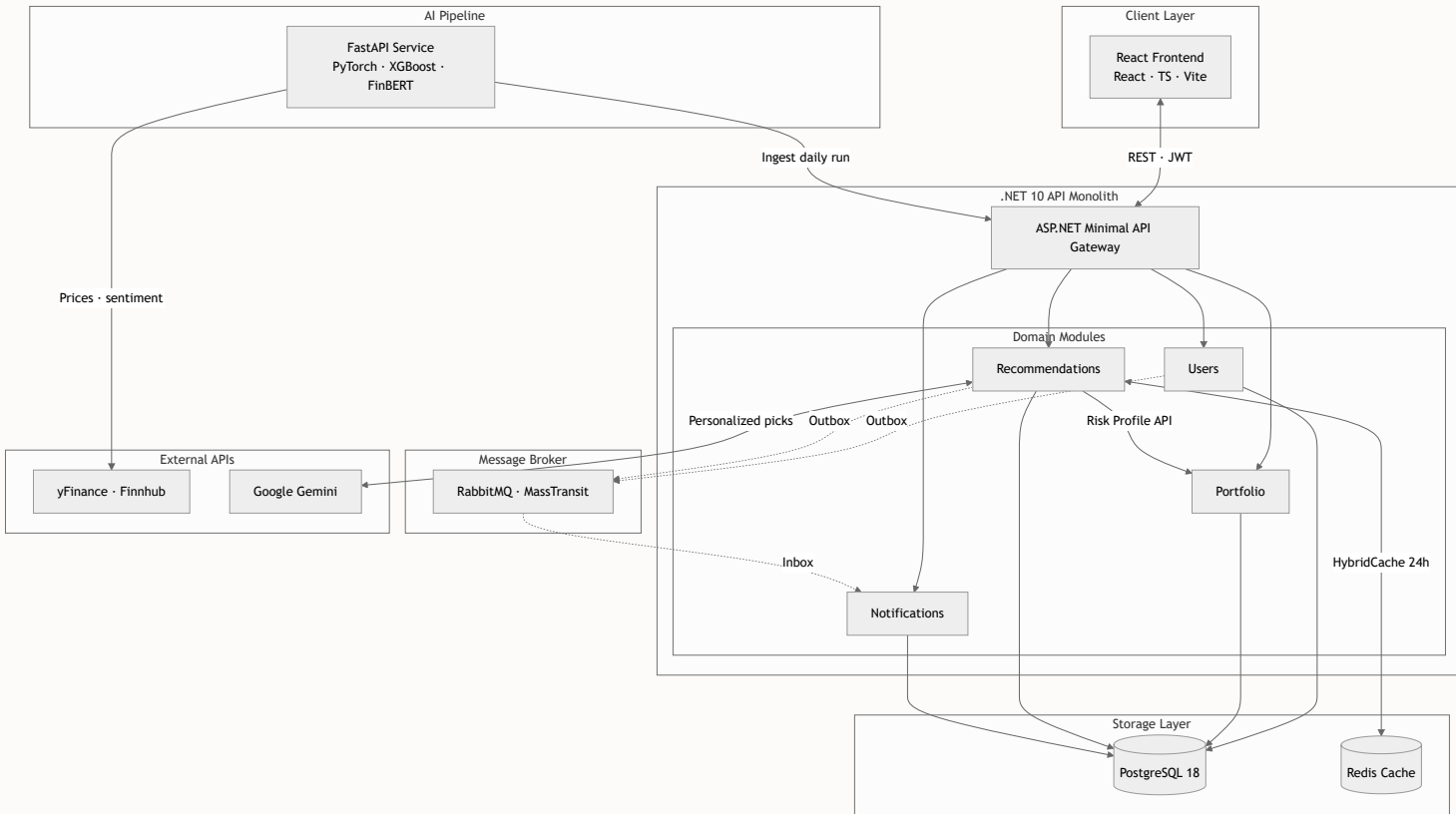
SAMPLE — AAPL, DAILY OHLCV · 2024 (SPLIT-ADJUSTED)

DATE	OPEN	HIGH	LOW	CLOSE	VOL
2024-06-03	191.24	193.32	190.87	192.36	50.1M
2024-06-04	192.97	193.64	191.37	192.68	47.5M
2024-06-05	193.72	195.21	193.20	194.19	54.2M
2024-06-06	194.01	194.81	192.50	192.81	41.2M

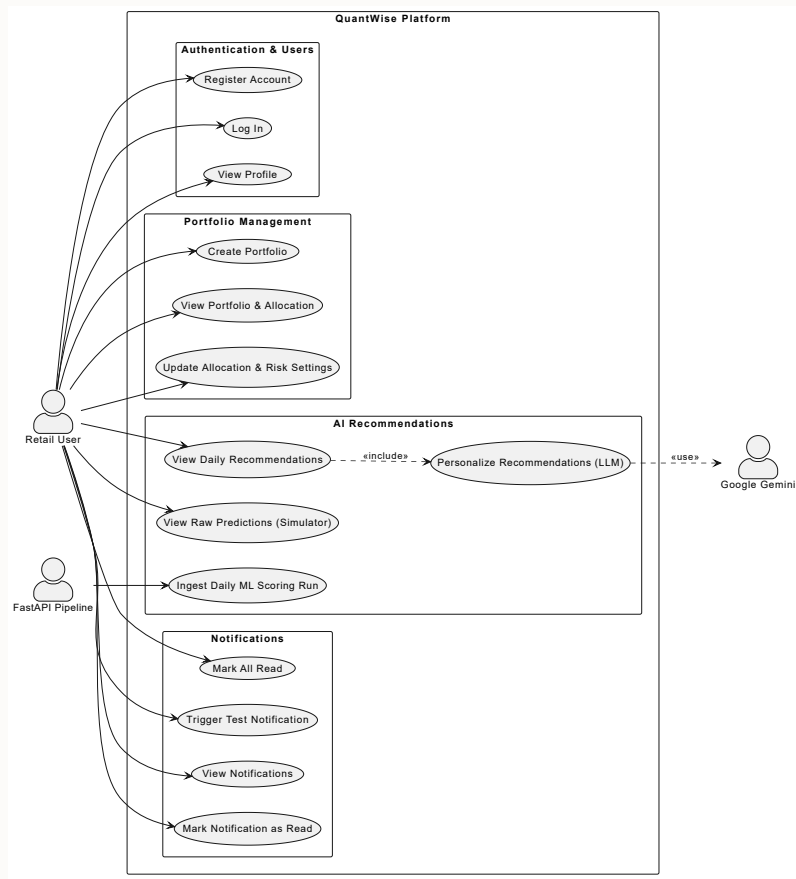
TARGET DISTRIBUTION — 30-DAY RETURN, Z-NORMALISED



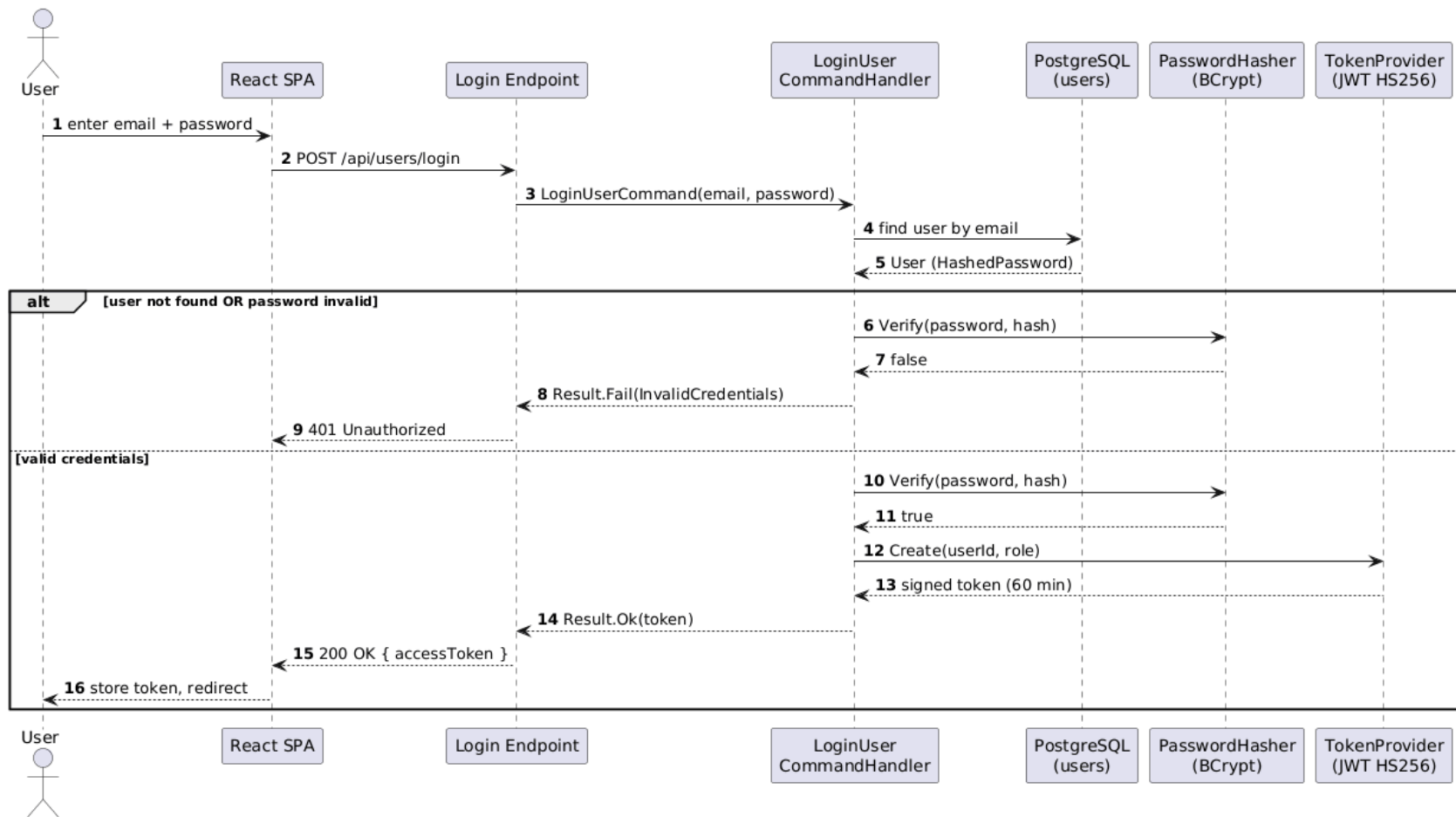
Container view — modular monolith + AI pipeline



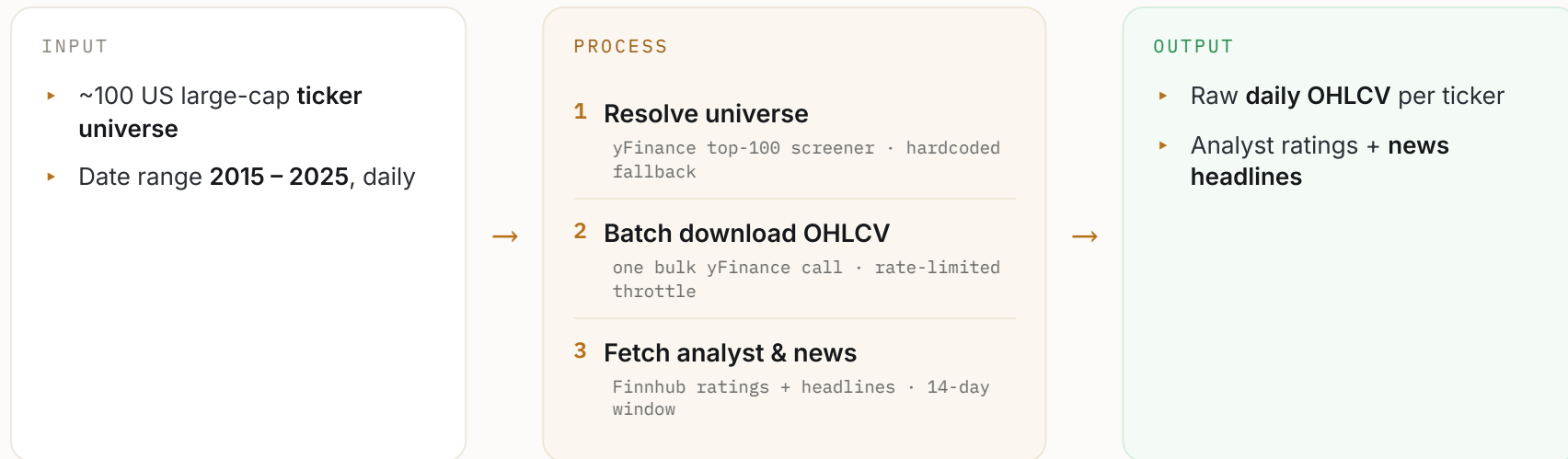
Use-Case Diagram



DESIGN · SEQUENCE — LOGIN (JWT)

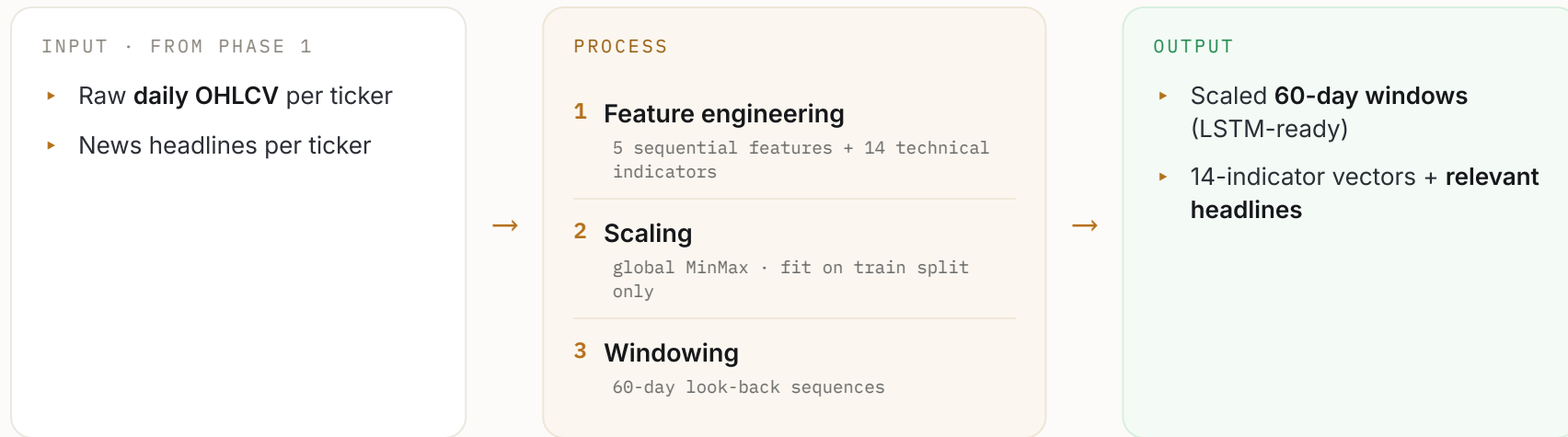


Phase 1 — Data Acquisition



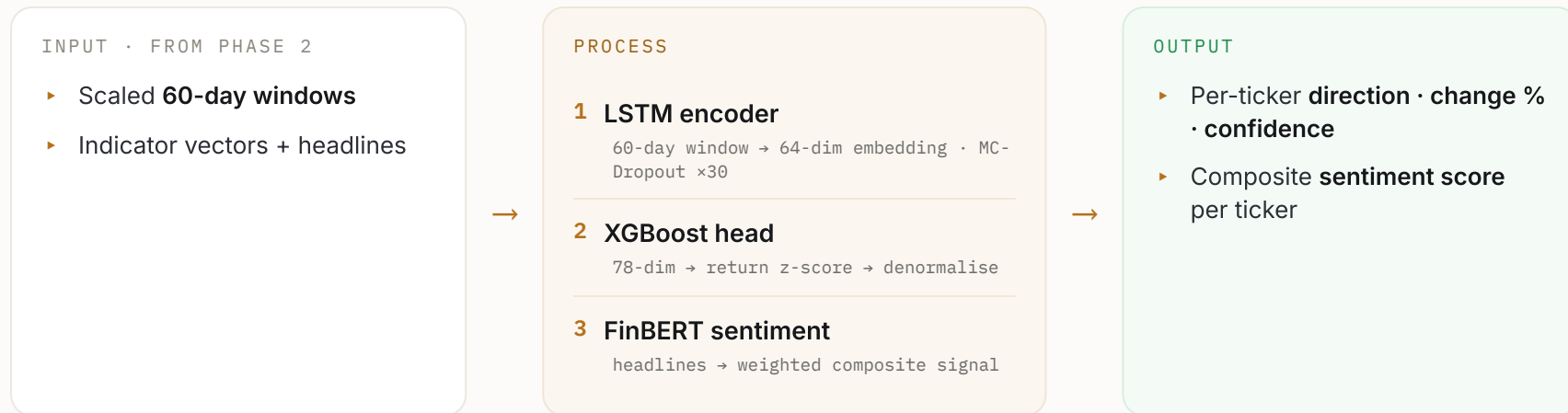
→ Output becomes the input to **Phase 2 · Pre-processing**

Phase 2 — Pre-processing



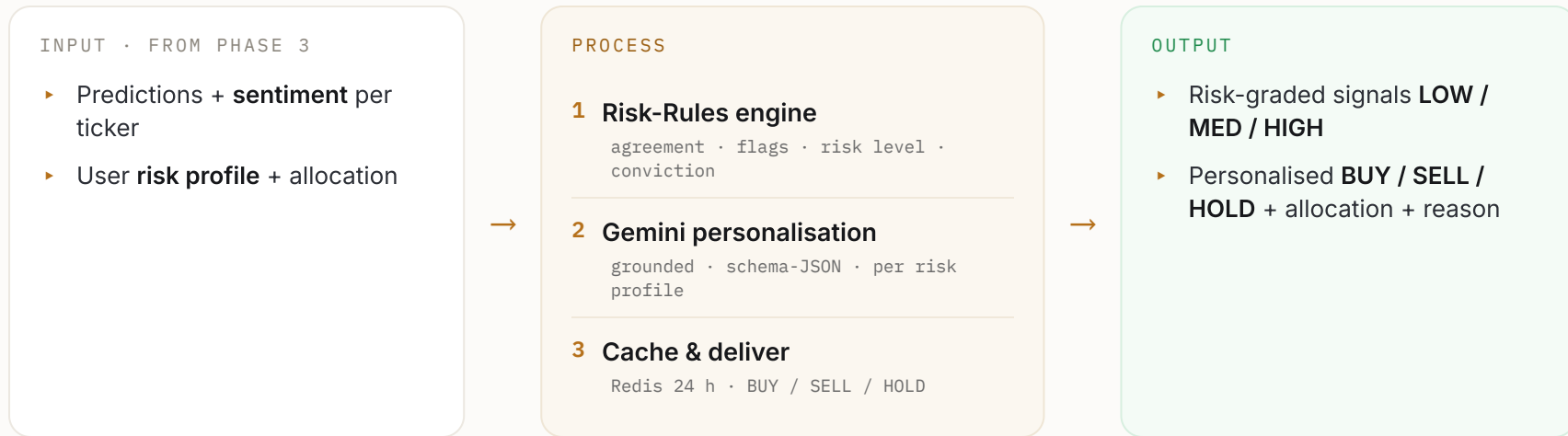
→ Output becomes the input to **Phase 3 · Model**

Phase 3 — Hybrid Model



→ Output becomes the input to **Phase 4 · Decision**

Phase 4 — Risk Grading & Decision



→ Delivered to the **React dashboard** — the user's personalised recommendation

What the pipeline produces — and how it's weighted

SAMPLE OUTPUT · 1 OF 100 RECORDS · RUN 2026-06-17

```
ticker: "MSFT"
direction: "UP" · change_pct: +7.66
confidence: 0.72 · signal: "POSITIVE"
sentiment_score: 0.54
analyst_rating: 4.27 (Buy) · pt_upside: +42.6%
agreement: "CONFIRMED"

risk_level: LOW conviction: 0.72

risk_flags: ["signal_confirmed"]
```

WEIGHTS BEHIND EACH PHASE

① Prediction → confidence

$\sqrt{(\text{signal_strength} \times \text{stability})} \times \text{data_quality}$

stability from MC-Dropout $\times 30$ · data_quality = share of inputs in range

② Sentiment → composite score

Consensus		40%
News		25%
Price target		20%
Up/downgrades		15%

③ Risk → conviction

$0.5 \cdot \text{confidence} + 0.3 \cdot |\text{sentiment}| \pm 0.2$

+0.2 if quant & sentiment agree · -0.2 if they contradict

Grounding the generative model

CONSTRAINED SYNTHESISER, NOT A PREDICTOR

- ✓ Uses **only the day's risk-graded data** — never invents tickers, prices, or numbers.
- ✓ Respects the grading — **no HIGH-risk or contradicted picks** for Conservative users.
- ✓ Output forced to **schema-valid JSON** (native responseSchema) · **3x retry** on parse fail.
- ✓ Allocations must **sum to 100%**; result **cached 24 h** per user in Redis.

ILLUSTRATIVE OUTPUT · { SUMMARY, PICKS[] }

```
ticker: "MSFT" · action: BUY
allocation_pct: 18
reason:
"Confirmed UP signal with strong analyst
backing and +42% target upside."
risk_note: "LOW risk · core holding."
fit: "Anchors a Moderate portfolio."
```

Why it matters: the model synthesises and explains pre-validated signals — sidestepping the hallucination risk of asking an LLM to forecast from scratch.

Hybrid vs standalone baselines

30-DAY TEST RMSE — LOWER IS BETTER

Standalone LSTM ≈ 0.285



Hybrid · LSTM → XGBoost 0.0949



Hybrid error is **≈ one-third** of the standalone LSTM, and **matches or surpasses the XGBoost-only** baseline across the majority of stocks.

DIRECTIONAL ACCURACY

Rises with horizon — **97.6%** at 365 days.

But that largely tracks the high **base rate of positive long-horizon returns** — the model adds little over an always-up predictor at long horizons, so the **30-day RMSE is the more honest signal**.

Mostafa, Ahmed, Darwish & Solayman (2026) — A Hybrid LSTM-XGBoost Framework · 14 US equities, multi-horizon. Deployed QuantWise model = single 30-day variant.

PUBLICATION



ACCEPTED • PENDING PUBLICATION

"A Hybrid LSTM–XGBoost Framework for Multi-Horizon Stock Return Prediction Across Diversified Equity Portfolios"

Accepted at IMSA • IEEE Egypt Section

AUTHORS

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TECHNOLOGIES USED

The stack, by layer

FRONTEND



React



TypeScript



Vite

TanStack Query

Framer Motion

BACKEND · .NET



ASP.NET Core 10

MediatR · CQRS

EF Core

FluentValidation

JWT

MESSAGING & CACHE



RabbitMQ

MassTransit



Redis

ML PIPELINE



Python



FastAPI



PyTorch PyTorch · LSTM

XGBoost

FinBERT

DATA & AI



PostgreSQL 18

yFinance

Finnhub



Google Gemini

DEVOPS & TESTING



Docker Compose

xUnit

Testcontainers

Playwright

k6

FEATURE LIST

What QuantWise offers



Secure accounts

JWT login, hashed passwords.



Risk profiling

Questionnaire → Conservative / Moderate / Aggressive.



Personalised recommendations

BUY / SELL / HOLD + allocation + reasons.



Holdings-aware advice

HOLD or SELL on what you already own.



Portfolio & target mix

Allocation, amount, per-pick dollar split.



Notification centre

Welcome & daily-run alerts, unread badges.



Live market hub

Real-time quotes and symbol search.



Learning environment

Allocate across picks, see projected outcome.



Profile management

View and update account & risk profile.

How we tested the system

UNIT

xUnit · NSubstitute

CQRS handlers and domain rules in isolation, with substituted dependencies.

55 tests · all pass

API · INTEGRATION

xUnit + WebApplicationFactory

Hits the real API over real Postgres & Redis (Testcontainers); Swagger / Postman for manual checks.

13 tests · all pass

GUI · SYSTEM

Playwright

Drives the live React app end-to-end like a real user — the role Selenium plays elsewhere.

12 black-box cases · all pass

MODEL

RMSE · MAE · Dir. accuracy

Hybrid scored against standalone LSTM & XGBoost baselines (see Results).

reported & benchmarked

LOAD

k6

50 virtual users hammering the read paths for two minutes.

0 errors · p95 6.9 ms

COVERAGE

47.3% / 54.8% core

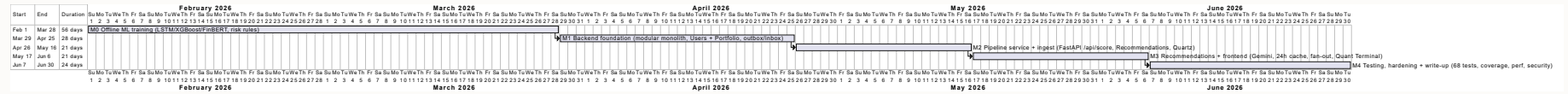
Line coverage merged across 6 projects via coverlet + ReportGenerator.

68 automated tests

Black-box test cases

ID	AREA	TEST	EXPECTED RESULT	STATUS
TC-01	Auth	Register a new account	201 · account created	PASS
TC-02	Auth	Login — valid / invalid	Token issued · 401 generic on bad creds	PASS
TC-03	Onboarding	Complete the questionnaire	Risk profile · allocation sums to 100	PASS
TC-04	Security	Read another user's data	HTTP 403 forbidden	PASS
TC-05	Recommendations	Open the dashboard	Personalised BUY picks + reasons	PASS
TC-06	Portfolio	View allocation in dollars	\$5,000 of \$10,000 shown per pick	PASS
TC-07	Notifications	Open the bell after registering	One unread "Welcome" notification	PASS
TC-09	Market	Open the Market page	Live quotes for 8 tickers	PASS
TC-11	Learning	Open the Learning view	Latest pipeline predictions shown	PASS

Project Schedule (Gantt)



Thank You

We welcome any questions.